

Shih-Ni Prim
North Carolina State University
December 7, 2025

Jeff McLean, Teaching Associate Professor, Director
Department of Statistics and Operations Research
University of North Carolina at Chapel Hill

Dear Dr. McLean and the Search Committee,

Currently a PhD candidate of the Department of Statistics at North Carolina State University, I am writing to express my qualifications and interest in the position of Teaching Assistant Professor at the department of Statistics and Operations Research. I have experience in teaching as a teaching assistant, an instructor, and a mentor. My research spans a few different directions and has prepared me for teaching and design courses in a wide variety of topics.

My teaching experience has been in a variety of settings. I served as a teaching assistant in two master's level statistics courses (ST 501 & ST 503—Fundamentals of Statistical Inference I & Fundamentals of Linear Models and Regression), a guest lecturer for ST 421 (Introduction to mathematical statistics I). I helped the students learn the process of learning and emphasized the importance of logical and critical thinking. In Summer 2025, advising six undergraduate students from NC State and two high school students from NC School of Science and Mathematics on data science projects. The one-on-one advising experience provided me with a sneak peek of how to mentor junior researchers. In Spring 2026, I will serve as a section instructor for ST 311 (Introduction to Statistics) and build on my teaching experience in the classroom. While these experiences were from my current role as a PhD student, I have prior teaching experience as the sole instructor; as an Adjunct Assistant Professor, I designed and taught MUS 101, Introduction to Western Music at Wake Forest University (WFU) in Fall 2017. My teaching and advising experience helped me form the teaching philosophy of emphasizing critical thinking and the ability to learn over learning details.

During my PhD study in statistics, I take full advantage of the wide range of faculty expertise offered by my department at NC State University and work on three research projects in different directions broadly connected by the concept centering uncertainty quantification (UQ). My first project, with Dr. Brian Reich, is about spatial and causal inference; we developed a fully Bayesian, spectral method to account for unmeasured spatial confounding and provide unbiased effect estimates and conservative credible intervals for multiple exposures and multiple outcomes. Originating from a summer internship at the Lawrence Livermore National Laboratory, my second project concerns surrogate modeling for computer experiments. With Dr. Kevin Quinlan (LLNL) and Dr. Annie Booth (Virginia Tech), I developed an active learning method that utilizes Gaussian processes as surrogate models and joint predictive probabilities as

acquisition functions to efficiently find sampling points to locate the optimal design for contour locations from multiple computer experiments. After two projects motivated by applications, I turned to fundamentals and am currently applying possibilistic inferential models (IM) to decision theory with Dr. Ryan Martin and Dr. Jonathan Williams. Using possibility measures, IM incorporates imprecision and provides validity guarantees. These projects provide breadth in thinking and wide opportunities for current and future collaboration. After graduation, I plan to continue some of these directions, as detailed in my research statement. Though the position's main responsibility is to teach, my research experience has prepared me to teach and develop new courses in a wide range of topics.

Being a former Tar Heel, I cannot be more excited to see the prospect of returning to UNC Chapel Hill, a vibrant campus in a beautiful city. I am looking forward to the opportunity to work with students through teaching and advising. Should you have any questions, please feel free to reach out. I am looking forward to hearing from you.

Sincerely,

Shih-Ni Prim

sprim@ncsu.edu

(336) 254-3431